

Xiuyin Li

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Research assistant roles — immediate availability; one-year minimum | IELTS: 7.0 (W: 7.0, S: 6.5)

Reliable LLM agents: systems, empirical evaluation, and failure analysis for long-horizon interaction

Last updated: September 2, 2026



EDUCATION

The University of Hong Kong Master of Data Science **3.51/4.30** Sep 2025 – Jul 2027 (*expected*)

Deep Learning (A+), NLP and Text Analytics (A+), Data Mining (A-), Statistical Inference (A-), Advanced Statistical Modelling (A-), Programming for Data Science (A)

Coursework completed; practicum remaining; degree requirements expected in December 2026.

Shandong University B.S. in Statistics **86.52/100** Sep 2021 – Jun 2025

Machine Learning (95), Statistical Computation (99), Bayesian Statistics (96), Mathematical Statistics (92), Multivariate Analysis (91), Python Programming (95), Database Technologies (99), Mathematical Modeling (90)

Minor in Computer Science and Technology **88.86/100** Sep 2022 – Jun 2025

Data Structures (87), Operating Systems (91), Software Engineering (92), Object-Oriented Technology (90)

RESEARCH EXPERIENCE

B.S. Thesis: EM and MCEM for Parameter Estimation in Regression Models May 2025

- Derived and implemented EM/MCEM in R for Student's t regression and linear mixed models, including ECME and REML updates; validated in 200- and 500-replication simulations and against standard R packages.
- Reworked computations to avoid dense matrices for 8,000 UTKFace embeddings and 59,699 CIFAR-10H trials; the t model outperformed a Gaussian baseline in-sample, while the mixed model matched `lme4`.

Synthetic Control Methods for Causal Inference Research Collaboration · Jul–Aug 2024

- Explored synthetic-control applications and extensions with Yunhao Wang and three other undergraduate students through reading and reproduction of R studies spanning design and treatment-effect estimation.

SELECTED PROJECTS

Meeting Agent — Multimodal Hybrid RAG with Long-Term Memory Project Lead · Apr 2026

- Designed and implemented a hierarchical hybrid RAG over meeting transcripts, parsed text, OCR, and image captions, combining Chroma retrieval with SQLite FTS5/BM25, RRF, and semantic reranking.
- Evaluated retrieval with Recall, MRR, and nDCG; used LLM-as-judge scores for faithfulness, relevance, and context precision; tested memory persistence, ingestion, streaming, and failure recovery.
- Added cross-session memory with LLM-extracted facts, semantic retrieval, TTL/importance decay, alias-aware entities, and episodic summaries for persistent recall across meetings.

WebAgent — Browser Agent & Evaluation Harness Project Lead · Nov 2025–Present

- Built a model-agnostic Chromium agent with screenshot-and-DOM grounding, policy-gated typed tools, atomic checkpoint recovery, and caption-grounded PDF analysis across hosted APIs and local vLLM.
- Developed versioned traces and a 36-task harness across public web, sandbox, and forced resume, with independent judging and confidence calibration; on one frozen date, both models passed 35/36.

Zero-Shot CNNs from Textual Descriptions – Reprod. & Ext. Project Lead · Mar 2026

- Implemented the core Ba et al. (ICCV 2015) zero-shot CNN predictors in PyTorch from scratch, generating class-specific FC/conv weights from text for unseen classes under a paper-aligned seen/unseen protocol.
- Ran paper-aligned CUB/Flowers evaluation plus 13 extensions (3 losses, 4 SBERT/CLIP variants, 6 backbones); with joint/VGG-19/BCE fixed, multi-granularity CLIP achieved unseen-class ROC-AUC 0.680→0.886.

TEACHING

Teaching Assistant, Shandong University — Mathematical Analysis I & II, PDE Sep 2024 – Jun 2025

Graded homework and answered questions; translated PDE slides and assisted with calculus-textbook LaTeX.

SKILLS

Machine Learning & AI: PyTorch, scikit-learn; Hugging Face Transformers, PEFT/LoRA, vLLM

Agents & Evaluation: LangChain, ReAct, MCP, Playwright/CDP, Pydantic, agent orchestration, offline evaluation

Statistics & Modeling: Python, R, SQL; NumPy, pandas, SciPy; mixed-effects modeling

Software & Reproducibility: Git, Linux (CLI), Docker, pytest, CI/CD, conda/uv, $\LaTeX$; C++, Java